

Tony Lin

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EDUCATION

University of Virginia, School of Engineering and Applied Science

GPA: 3.99/4.0 — B.S. Computer Science and Math

Charlottesville, VA

Aug. 2024 – May 2028

Thomas Jefferson High School for Science and Technology

GPA: 4.394/4.0

Alexandria, VA

Aug. 2020 – June 2024

- **Award:** 2024 AIME Qualifier (American Invitational Mathematics Examination)

EXPERIENCE

Enterprise AI Intern

May 2026 – Aug. 2026

Humana

Remote

- **Owned** the design and implementation of an evaluation framework benchmarking Humana's HEDIS engine (a **LangGraph** pipeline extracting compliance evidence from clinical documents) against clinician-reviewed ground truth across **8 quality measures**
- **Developed** automated evals – recall/precision, **LLM-as-a-judge** scoring, confidence calibration (**ECE**, **Brier score**) – gating prompt and model changes before production promotion
- **Drove** prompt refinement for the OMW measure model, using error analysis and nurse feedback to raise extraction precision from **76% to 97%**

AI/ML Intern

May 2025 – Aug. 2025

National Institute of Standards and Technology

Hybrid

- **Engineered** a RAG pipeline (LangChain, ChromaDB) benchmarking BM25, cosine similarity, and ensemble retrieval – **boosting** accuracy 12% and **cutting** latency 30% across 4k+ knowledge base articles
- **Automated** ingestion of 10k+ multimodal documents via Docling and VLMs, **reducing** manual processing effort by 80%

Machine Learning Evaluation Intern

Nov. 2024 – May 2025

ML@UVA

Remote

- **Partnered** with 6 teammates and client LMI to benchmark GraphRAG (BLEU, LLM-as-a-judge, cost, latency) – findings **adopted** for client pilot trials
- **Benchmarked** RAPTOR, CRAG, Self-RAG, and LightRAG against vanilla RAG on a legal dataset – found full GraphRAG's 10x cost and 100x latency outweighed its marginal gains, while LightRAG delivered the best quality/cost tradeoff

PROJECTS

TDILEARNED | Next.js, React Flow, FastAPI, LangGraph, Supabase

Aug. 2026 – Present

- **Built** an agentic discovery engine – an infinite **React Flow** canvas that expands any topic into a spatial mindmap via a multi-agent **LangGraph** pipeline (planner, parallel search workers, source verifier, monograph synthesizer) querying DuckDuckGo, Wikipedia, Wikimedia, and OpenStreetMap
- **Built** a **FastAPI** backend streaming live node expansion and token-level chat over **SSE** (95ms time-to-first-token), validating model output against **Pydantic** schemas with retry fallbacks, routed through **Cerebras CS-3** (3k tok/s) with Mistral fallback
- **Precached** 886+ topic hubs via two-tier disk+memory caching in **Supabase** for instant topic exploration; deployed on **Vercel** + **Azure App Service** with a **Cloudflare Workers** edge media proxy

Dime | LangGraph, Agno, FastAPI, NextJS

May 2025 – July 2025

- **Built** an NBA analytics agent (**LangGraph**) answering natural-language queries for player/team stats and live game data, streamed through a **FastAPI** + **Pydantic** SSE backend to a Next.js dashboard
- **Implemented** two-tier memory+disk caching that eliminated external API rate-limit failures

Snaek's Value List | Supabase, Azure Web App, Discord.js, Netlify, Cloudflare

Feb. 2026 – Apr. 2026

- **Built and shipped** a value-tracking platform (trade-ad posting, calculator) for a Counter Blox (Roblox) trading community, reaching **1,000 monthly users** and replacing a manual spreadsheet as the source of truth
- **Migrated** item data (names, images, values) from a wiki scrape and legacy spreadsheet into a structured **Supabase** database; served images via **Cloudflare** CDN to keep hosting costs free at scale
- **Built** a role-based admin panel for managing item values, plus a companion **Discord bot** querying the same backend for in-server lookups
- **Integrated** a VLM-based inventory parser (Qwen3.6-27B via Groq) letting users upload a screenshot of their inventory and auto-populate item values, cutting manual entry and reaching **90% parsing accuracy**

SKILLS

Programming Languages: Python, TypeScript, JavaScript, Java

Frameworks & Libraries: React, Next.js, React Flow, Tailwind CSS, LangChain, LangGraph, Agno, FastAPI, Pydantic

Platforms & Tools: Git, GitHub, Supabase, PostgreSQL, Cloudflare Workers, Azure, Docker, Databricks, Langfuse